QUENCH.

QUANTIFIED EVIDENCE NAVIGATOR FOR COMBUSTION HAZARDS

NASA plans to send crews to the Moon in cabin air with **34 % oxygen**. No solid material has ever been flight-tested above **22.2 %**.

22.2 %
highest oxygen in any solid-material flight test

34 %
oxygen in the exploration atmosphere

0
solid or liquid test runs in that atmosphere

986
flight test runs from 24 NASA investigations

01

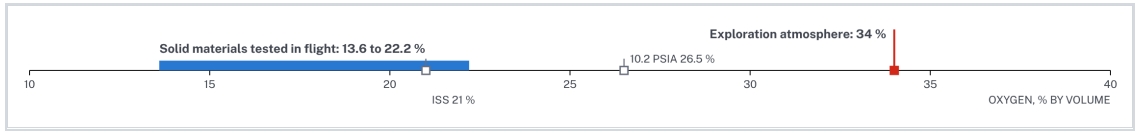
Decades of fire experiments in orbit. Do they cover the air a Moon base will use?

We read every combustion investigation in NASA's PSI archive: **24 investigations, 986 test runs**, 352 papers and 6.3 TB of raw data, and placed each run by the oxygen and pressure it burned in.

> “Because of the pure oxygen atmosphere, the fire spread quickly…”
>
> Apollo 1 · NASA Office of Safety and Mission Assurance

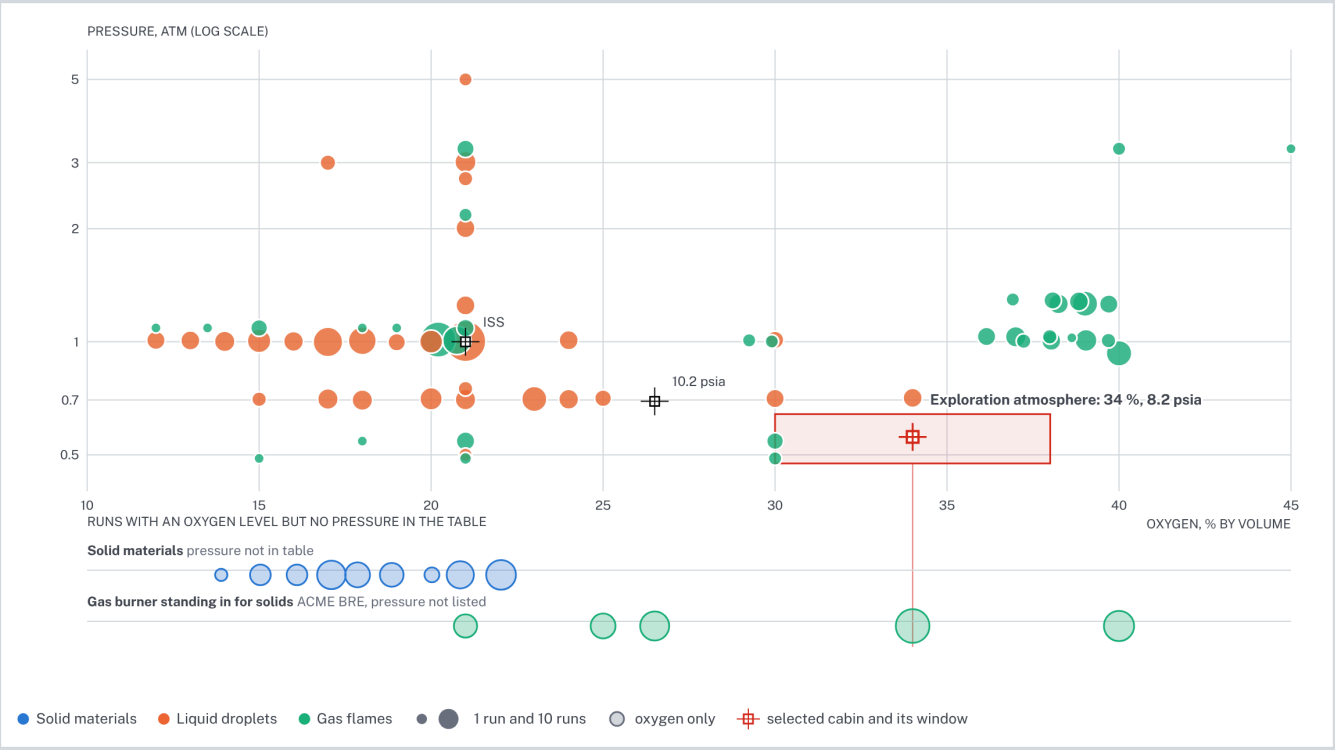
Oxygen-rich cabins have killed before. The exploration atmosphere raises oxygen again, to 34 %.





02

The evidence stops at 22.2 % oxygen for every solid material

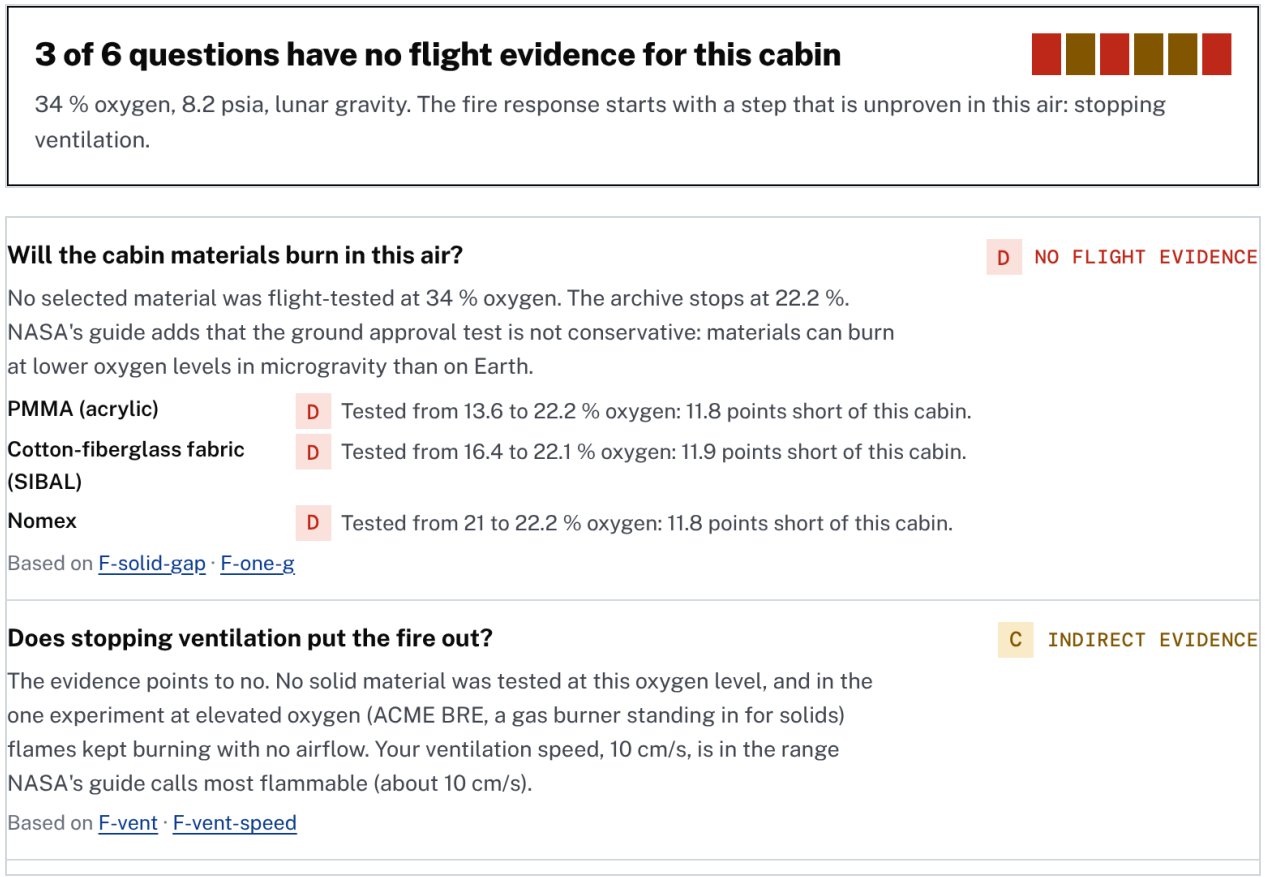


WHAT BURNED	SPACE STATION 14.7 PSIA · 21 %	REDUCED-PRESSURE CABIN 10.2 PSIA · 26.5 %	EXPLORATION ATMOSPHERE 8.2 PSIA · 34 %
SOLID MATERIALS	108 Tested	0 No flight tests	0 No flight tests
LIQUID DROPLETS	147 Tested	43 Tested	0 No flight tests
GAS FLAMES	80 Tested	0 No flight tests	9 Thin evidence

- Droplets reached 34 %, but at **0.70 atm**, not 0.56.
- At 25 % oxygen and above only **8 %** of droplet flames went out on their own; below 18 % it was 92 %.
- Every run was in orbit. None at lunar gravity.

03

Mission check: describe a cabin, get graded evidence and the tests still needed

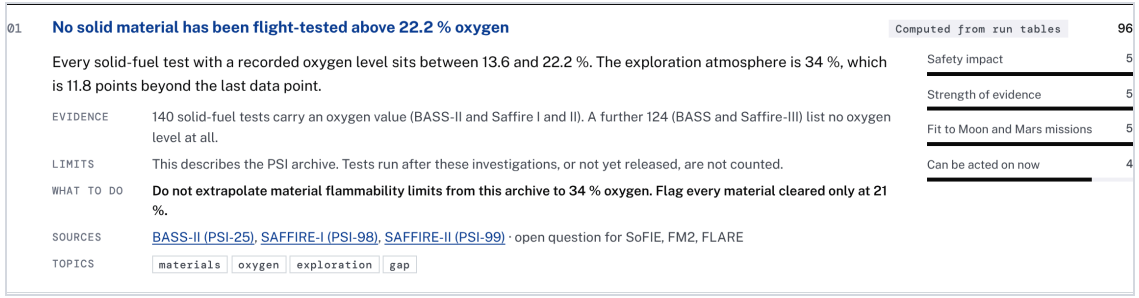


Oxygen, pressure, gravity, ventilation, materials, extinguishing agent. For each fire-safety question QUENCH grades the flight evidence **A to D** by written rules, shows the runs it used, and ranks what to test next.

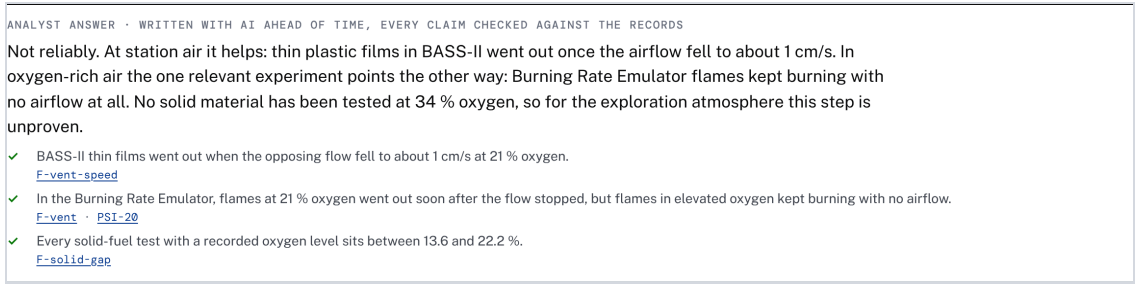
- Lunar habitat: **3 of 6 questions have no flight evidence.**
- Space station: evidence for every question.
- One click copies the report.

04

15 ranked findings, and an analyst that answers only from NASA's records



- Each finding: evidence, limits, action, sources. You set the weights.
- NASA's own results: the material approval test is **not the worst case**.



- Runs in the browser: **no API, no key, no cost**.
- 30 answers written with AI ahead of time; code checked all **56 claims** against the records.

05

Lab: a model that knows its limits, and results read from raw NASA video

CHANCE THE FLAME GOES OUT ON ITS OWN

<1 %

very unlikely for a 3 mm heptane droplet at 34 % oxygen and 8.2 psia

× Outside the tested conditions: the model is extrapolating

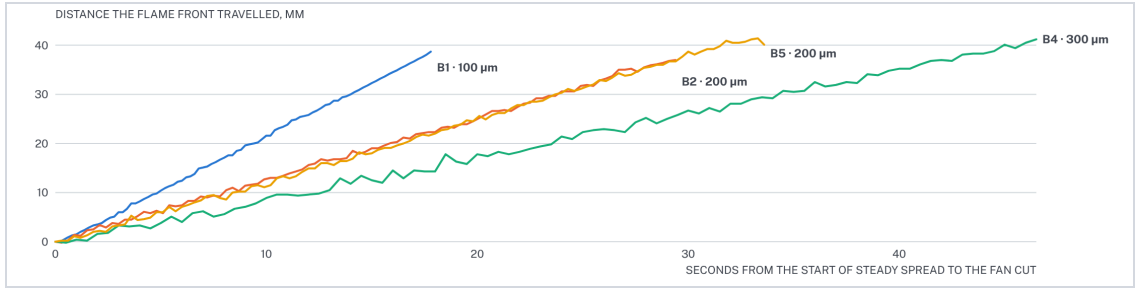
Beyond the tested range of pressure: FLEX ran 10.2 to 44.9 psia. Read the number as a direction, not a forecast.

Burning-rate constant if it keeps burning: about **0.73 mm²/s** (typical model error ±0.09), so a 3 mm droplet would last about 12 s by the d^2 law.

82 %
right on unseen test days

69 %
always guessing

Trained on **261 FLEX runs**. It flags extrapolation.



5 / 5
independent checks pass

1.6 %
between repeat tests

Flame out **1.9 s** after the fan cut; NASA's report says within 2 s.

06

Built to be trusted, free to run

TRUST

- One script recomputes every number from NASA's raw tables.
- Every quote is matched word for word against its NASA source, or the build stops.
- **34 automated tests** recount key facts with independent code.
- No model runs at use time, so nothing can be made up on the spot.

NEXT

- Feed the gaps to SoFIE and FM2, which can test elevated oxygen and lunar gravity.
- Run the video tracker over all 244 BASS-II videos and publish the results table the archive lacks.
- On Earth, the same science already made firefighter air tanks lighter and detectors smarter.